EDOS-kilo

A full screen editor for ELF-DOS, written in 1802 Assembly language. It is a port of the Elf/OS kilo editor by Gaston Williams, which is based loosely on the Kilo editor by Salvatore Sanfilippo aka antirez and the Elfos-Edit editor by Mike Riley.

Platform

The ELF-DOS kilo editor runs on an 1802 based Microcomputer running the ELF-DOS operating system, with an ANSI (VT100 style) terminal as the console. ELF-DOS runs on the same hardware as Elf/OS, such as the Pico/Elf by Mike Riley or the 1802-Mini by David Madole.

The editor takes the screen size from ELF-DOS. Run `TERMSIZE` first (a line in `AUTOEXEC.BAT` will do) so that ELF-DOS knows the real size of the terminal. Without it the editor assumes a screen of 24 rows by 80 columns. The widest screen the editor uses is 132 columns.

Usage

```
kilo filename
```

The file name may include a drive letter and a path, such as `D:/src/hello.asm`. If the file does not exist, the editor starts with an empty buffer and creates the file when it is saved. The exit code is 0, or 1 if the editor could not start.

Building

The editor is assembled and linked with the Asm/02 assembler and Link/02 linker, the same versions used to build ELF-DOS. The Makefile expects them in `/opt/elfc`; set `ASM` and `LINK` to use them from somewhere else.

```
make          # assemble src/*.asm and link ./kilo
make clean
```

Copy the `kilo` file that this produces into the `/bin` directory of an ELF-DOS volume.

`elfdos-sdk/` is a copy of the ELF-DOS SDK that the editor is built against; see `elfdos-sdk/MANIFEST.txt` for the ELF-DOS version it came from, and `elfdos-sdk/DEVELOPER_GUIDE.md` for the program interface. Rebuild the editor against a newer SDK if the ELF-DOS kernel's minor version changes.

Differences from the Elf/OS version

- File names can be a path of up to 79 characters on the command line. A new name entered with the Rename function can have up to 64 characters, and may contain a drive letter, a path and hyphens.

- Keys are always read through the ELF-DOS kernel, so the editor works with whatever console ELF-DOS is using. The circular key buffer for the hardware UART has been removed.
- Spill files are created in the current directory of the current drive.
- A save that fails part way through a large file is reported as an error.

Kilo Key Commands

Keys	Command
Ctrl+J, Left	Move cursor Left
Ctrl+K, Right	Move cursor Right
Ctrl+U, Up	Move cursor Up
Ctrl+N, Down	Move cursor Down
Home	Move cursor to the beginning of line
End	Move to End of line
PgUp	Page up to previous screen
PgDn	Page down to next screen
Insert	Toggle insert/overwrite mode
Ctrl+D, Delete	Delete character
Ctrl+H, Backspace	Move back and delete character
Ctrl+I, Tab	Move forward to next tab stop
Ctrl+], Shift+Tab	Move back to previous tab stop
Ctrl+X	Exit, confirm if file changed
Ctrl+M, Enter	Insert Mode: insert new line or split at cursor
	Overwrite Mode: Move down to the next line.
Ctrl+?, Ctrl-^ (See Note)	Show help information

Keys	Function Set
Ctrl+F	File Functions
Ctrl+P	Page Functions
Ctrl+L	Line Functions
Ctrl+C	Cursor Functions

Note: On some older terminals the Ctrl+_ key combination replaces the Ctrl+? combination key for help. Some emulators may support one key combination, or the other, or both.

Kilo File Functions

	Keys	Function
Ctrl-F	S	Save file
	R	Rename and save file
	Q	Save file and quit
	X	Exit, confirm if file changed

Kilo Page Functions

	Keys	Function
Ctrl-P	U	page Up to previous screen
	D	page Down to next screen
	B	Move to Bottom of buffer
	T	Move to Top of buffer
	R	Refresh page on screen

Kilo Line Functions

	Keys	Function
Ctrl-L	C	Copy line to clip board
	X	Cut line into clip board
	V	Paste line from clip board
	D	Delete line
	J	Join lines
	S	Split line at cursor

Kilo Cursor Functions

	Keys	Function
Ctrl-L	H	Move cursor to the beginning of line (Home)
	E	Move to End of line
	W	show Where cursor is located in file (See Note)
	G	Go to line number (See Note)
	F	Find text
	I	Toggle insert/overwrite mode

Note: The line numbers are calculated based on 96 rows per buffer. If a buffer has lines added or deleted, so that its number of rows is not 96, then calculated line number in following buffers will be off by the difference.

Kilo Text Modes

- **Insert Mode**
    - Characters are inserted into the text
    - Tab inserts spaces into the text
    - Enter inserts a new line at the beginning or end of a line
    - Enter splits a line at the cursor inside a line
    - Delete at the end of a line joins the current line with next line
    - Backspace at the beginning of an empty line deletes the line
    - Line Split (Ctrl-L,S) will also insert lines or split a line
    - Line Join (Ctrl-L,J) will join lines
- **Overwrite Mode**
    - Characters overwrite existing characters in the text
    - Tab moves the cursor to the next tab stop position
    - Enter moves down to the beginning of the next line
    - Delete and backspace do not delete empty lines automatically
    - Line Split (Ctrl-L,S) at the beginning or end of a line will insert an empty line
    - Line Split (Ctrl-L,S) will split a line at the cursor inside a line
    - Line Join (Ctrl-L,J) will join lines
    - Line Delete (Ctrl-L,D) will delete a line

Copy, Cut & Paste

- Line Copy (Ctrl-L,C) will copy a line into the clip board
- Line Cut (Ctrl-L,X) will delete a line and copy into the clip board
- Line Paste (Ctrl-L,V) will paste a line from the clip board and insert it above the current line

Text Limits

- Lines can have up to 124 characters per line
- Tabs are defined as 4 character tab stops (1, 5, 9, etc.)
- Tab characters in a file are replaced by spaces up to the next tab stop when the file is loaded
- Other control characters in a file are removed when the file is loaded

File Limits

- Buffers are used to edit files larger than 96 lines
- Each buffer has up to 96 lines
- A file can have up to 255 buffers
- Buffers are saved to temporary spill files, while the file is edited
- Spill files are created automatically when a large file is loaded
- Spill files are named __*kilo.nn*, where nn is the 2 digit hex value of their index

- Spill files are deleted automatically when the application ends

Key Character Sequences

These key character sequences follow the VT102 terminal specification. In the table below, {ESC} is the Escape control character, ASCII 27, hex value $1B. Should one of these keys not behave as expected, you may need to configure your terminal program to return the expected key character sequence. Note that the Delete key can have either one of two possible sequences.

Key	Character Sequence
Up (Arrow)	{ESC}[A
	{ESC}A (See Note)
Down (Arrow)	{ESC}[B
	{ESC}B (See Note)
Right (Arrow)	{ESC}[C
	{ESC}C (See Note)
Left (Arrow)	{ESC}[D
	{ESC}D (See Note)
Home	{ESC}[1~
Insert	{ESC}[2~
Delete	{ESC}[3~
	DEL ($7F)
End	{ESC}[4~
PgUp	{ESC}[5~
PgDn	{ESC}[6~
Backspace	Ctrl+H ($08)
Tab	Ctrl+I ($09)
Shift+Tab	{ESC}[Z
Enter	Ctrl+M ($0D)

Note: On some older DEC video terminals, the arrow keys send {ESC}A through {ESC}D instead of the ANSI sequences {ESC}[A through {ESC}[D.

Repository Contents

- ***/src/*** -- Source files for the ELF-DOS kilo editor.
  - kilo.asm - Main assembly source file.
  - kilo_file.asm - Assembly source file for file routines.

    - kilo_spill.asm - Assembly source file for spill file routines.
    - kilo_dos.asm - Assembly source file for calls to the ELF-DOS kernel file functions.
    - kilo_data.asm - Assembly source file data and buffers.
    - kilo_util.asm - Assembly source file for utility routines.
    - kilo_buf.asm - Assembly source file for buffer utility routines.
    - kilo_cfg.asm - Assembly source file for configuration utility routines.
    - kilo_prmpt.asm - Assembly source file for prompt and functions and help information.
    - kilo_str.asm - Assembly source file for string and message functions.
    - kilo_keys.asm - Assembly source file for main key handler and save routines.
    - kilo_text.asm - Assembly source file for text key handler routines.
    - kilo_edit.asm - Assembly source file for text editing key handler routines.
    - kilo_line.asm - Assembly source file for line editing key handler routines.
    - kilo_move.asm - Assembly source file for cursor movement key handler routines.
    - kilo_input.asm - Assembly source file for input and confirmation functions.
    - kilo_dbg.inc - Assembly source file for optionally included debug functions.
- **/src/include/** -- Include files for the editor.
    - kilo_def.inc - Definitions for the kilo source files.
- **/elfdos-sdk/** -- The ELF-DOS SDK: kernel, BIOS and opcode definitions, library modules and the Developer's Guide.
- **Makefile** -- Builds the editor.

References

- ANSI Escape Sequences
- Wikipedia ANSI escape code
- VT102 User Guide
- VT102, VT100 and other DEC Video Terminals
- Build Your Own Text Editor
- Antirez Kilo Editor

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This code is based on a Elf/OS kernel written by Mike Riley and created with the Asm/02 assembler and Link/02 linker also written by Mike Riley.

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